Tables, Graphs & Constant Change

WARM UPS

TEACHER COPY | Display student pages; keep solutions private

1 | Solve

Solve for x. Show each step, then check your answer.

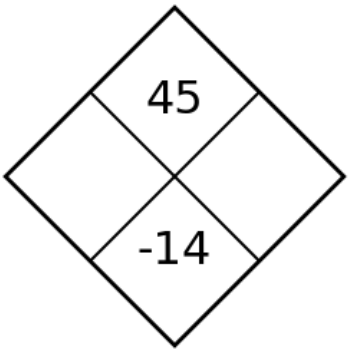
$$2(3x + 1) - 4 = 22$$

ANSWER / TEACHING NOTE

2(3x + 1) - 4 = 22 → 6x + 2 - 4 = 22 → 6x = 24 → x = 4.

2 | Diamond

Complete the diamond.



Tables, Graphs & Constant Change

WARM UPS

3 | Spiral Review

Is {(-1,4), (0,4), (2,4)} a function? Why?

> **ANSWER / TEACHING NOTE**
> Yes: each input has exactly one output; repeated outputs are allowed.

4 | Spiral Review

List the domain and range of {(2,7), (4,9), (6,7)}.

> **ANSWER / TEACHING NOTE**
> Domain {2,4,6}; range {7,9}.

Tables, Graphs & Constant Change

LEARNING FOCUS & NOTES

Learning focus | LG 02-ican-4 / LG 03-ican-1

- I can recognize constant additive change in a table.
- I can compare vertical change with horizontal change.
- I can connect a table, graph, and equation that describe the same linear pattern.

Vocabulary | Highlight the important words

Linear relationship

A relationship that has a constant rate of change and graphs as a straight line.

Constant difference

The same change in output for each equal change in input.

Horizontal change (Δx)

The difference between x-coordinates of two points; movement left or right.

Vertical change (Δy)

The difference between y-coordinates of two points; movement up or down.

Slope

The ratio of vertical change to horizontal change, Δy / Δx.

Starting value

The output at input zero; where a linear graph crosses the y-axis.

Meaning & Method | Rewritten review notes

1. Compare equal input intervals when checking for a linear pattern. A constant output difference is meaningful only when the input step is constant.

2. Compute slope as change in y divided by change in x. Keep the SAME point order in numerator and denominator.

3. For a linear rule y = mx + b, m is the slope and b is the output when x = 0.

These are original, concise review notes aligned to the course lessons; they are not a reproduction of the textbook pages.

Tables, Graphs & Constant Change

EXAMPLES & YOU TRY ITS

Example

Calculate Δx and Δy from the table, find the slope, and write the equation of the line.

x	y
0	2
2	8
4	14

Find Δx and Δy. Calculate the slope and write the linear equation.

> **ANSWER / TEACHING NOTE**
> Δx = +2, Δy = +6; m = 6/2 = 3; initial output b = 2.
> Equation y = 3x + 2.

You Try It

Use this table to find Δx and Δy, calculate the slope, and write a linear equation.

x	y
0	11
2	7
4	3

Find Δx, Δy, slope, and equation.

> **Show Δx, Δy, slope, and equation.**
> Δx = 2; Δy = -4; m = -4/2 = -2; initial output b = 11.
> Equation y = -2x + 11.

How can you verify that the rate of change stays constant?

> **TEACHING NOTE**
> Every +2 change in x has a -4 change in y, so the slope stays -2 throughout.

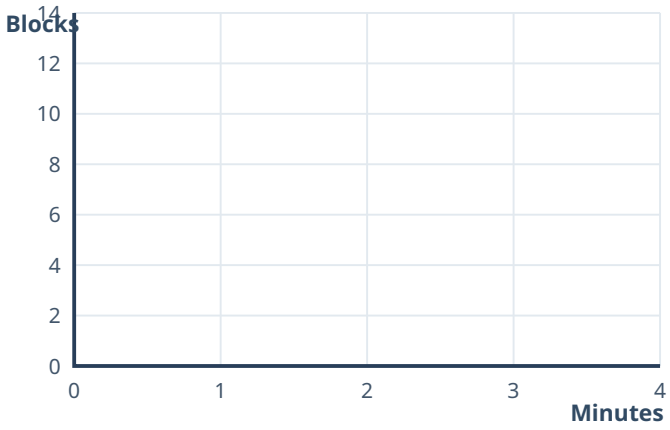
Tables, Graphs & Constant Change

LAUNCH

One story, three views

A walker starts 2 blocks from school and moves 3 more blocks away each minute. Complete the table and sketch the points; connect them if time is continuous.

Time (min)	Blocks from school
0	2
1	___
2	___
3	___



1 | How is the +3 visible in the table and on the graph? Write a rule using distance d and time t.

> **ANSWER / TEACHING NOTE**
> Table outputs: 2, 5, 8, 11. Graph these points. Rise = 3 for run = 1. Equation d = 3t + 2.

2 | A different walker starts 5 blocks away and moves 2 blocks each minute. Compare their changes for one minute (do not solve where they meet).

> **ANSWER / TEACHING NOTE**
> Walker 1 rate 3 blocks/min versus Walker 2 rate 2 blocks/min.

Partner share: Compare reasoning, not just answers. What would convince someone else your conclusion is correct?

Tables, Graphs & Constant Change

REVIEW & PREVIEW

1 Find the slope of a table with x-values 0,2,4,6 and y-values 6,12,18,24. Explain each change.

ANSWER / TEACHING NOTE

Δy = +6 for Δx = +2, so slope = 3.

2 A line passes through (0,6), (2,10), and (4,14). Find its slope and starting value.

ANSWER / TEACHING NOTE

Slope = (10 - 6)/(2 - 0) = 2; starting value = 6.

3 ★ The table (0,8), (1,5), (2,2), (3,-1) is linear. Find slope and write the rule.

ANSWER / TEACHING NOTE

m = -3 and b = 8, so y = -3x + 8.

Tables, Graphs & Constant Change

REVIEW & PREVIEW

4 ★ Compare: Pattern A adds 2 to the output for each +1 input. Pattern B adds 4. Which graph is steeper? Explain.

ANSWER / TEACHING NOTE

B is steeper: its positive slope 4 is greater than A's positive slope 2.

5 Spiral: Is {(1,2),(1,2),(3,5)} a function? State the domain and range.

ANSWER / TEACHING NOTE

Yes; repeating exactly (1,2) does not violate the function rule. Domain {1,3}; range {2,5}.

6 Preview: A runner gains 18 meters in 3 seconds. What is the average rate of change, including units?

ANSWER / TEACHING NOTE

18 m / 3 s = 6 m/s.

Evidence check: Star a question you can explain confidently. Circle one you want to revisit on Monday, November 9.